

Qatar-2b

R-0723

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250401 · created 2026-07-28T00:08:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460766.8155 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.022 +/- 0.064
Transit depth (Rp/Rs)^2	0.05 +/- 0.29 [%]
Orbital Inclination (inc)	89.8 +/- 1.73
Airmass coefficient 1 (a1)	0.42 +/- 0.24
Airmass coefficient 2 (a2)	0.61 +/- 0.98
Scatter in the residuals of the lightcurve fit is	57.47 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	4.75
Transit Duration (day)	0.0683 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.022 ± 0.064 vs 0.16327 (deviation 2.0σ)
Duration fit vs published	0.0683 d vs 0.0767 d (deviation 2.37σ)
Mid-time O–C	-2.6 min
Depth SNR	0.3
Residual scatter	57.47 %

Light curve

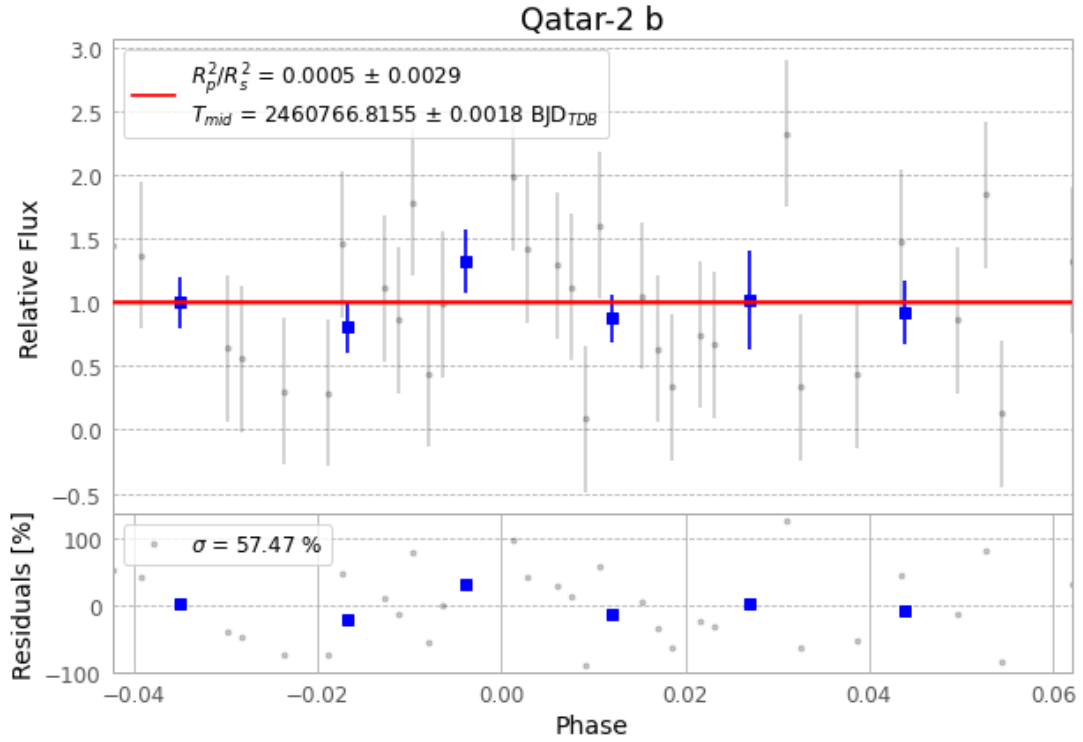


Figure 1 — FinalLightCurve_Qatar-2 b_2025-03-31.png (SHA-256 06f25ad621e22626...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [450, 414], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5e2964ddb175d17c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

68 FITS frames (45 MB), Qatar-2250401060317.FITS → Qatar-2250401092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250401.log (cc2ab58ef7b5...), FinalLightCurve_Qatar-2 b_2025-03-31.png (06f25ad621e2...), FinalLightCurve_Qatar-2 b_2025-03-31.csv (f6fdcafe4628...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:08Z.